

ADG classes: gapped k -mers

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Today we will implement the kernel function (or equivalently: calculate similarity score) for gapped k -mer Support Vector Machine. Have a look at the script `gapped_kmers.py` from Moodle. It considers three very short sequences (AAACCC, AAAAA and ACC), along with parameters $l = 3$ and $k = 2$ – exactly as in lecture slides and the original Ghandi et al., PLOS Computational Biology 2014 paper.

1. The first part of the script implements naive calculation of the kernel function (similarity score) explicitly by enumerating and counting all the gapped l -mers from the list of sequences S .
2. Implement a more efficient (and less naive) calculation of the kernel function, using only l -mers found in sequences from S (no explicit representation of gaps).
3. Compare the timing of both approaches, using the sequences from <http://www.beerlab.org/gkmsvm/downloads/ref.fa>, and parameters $l = 10$, $k = 7$. Hint: you may wish to consider only a subset of 20 sequences.
4. The last part of the script implements the calculation of the kernel function (similarity score) using l -mer tree data structure. First get to understand the data structures, and then compare the timing to the previous two approaches.
5. Some l -mer pairs do not contribute to the overall similarity scores and can be ignored – which ones, and when could the pruning occur?

6. To further speed up, introduce an optional parameter $m_{\max}$, which limits the maximum number of mismatches while DFS-traversing the l -mer tree data structure. By setting $m_{\max}$ smaller than $l - k$, we only consider l -mer pairs that have at most $m_{\max}$ number of mismatches. Try to reduce calculation by ignoring l -mer pairs which potentially contribute less to the overall similarity scores.